

BORONGAN, PhilippinesWMO: **985530**Lat: **11.660N**Lon: **125.442E**Elev: **3**StdP: **101.29**Time Zone: **8.00 (E08)**Period: **01-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	22.5	23.1	21.2	15.8	23.8	21.8	16.5	24.1	6.9	26.5	6.3	26.9	1.3	270	0.315

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	5.9	33.0	27.7	32.2	27.6	31.6	27.5	28.2	31.5	28.0	31.2	27.7	30.8	2.7	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.2	23.0	30.5	27.0	22.7	30.4	26.7	22.3	30.0	90.5	32.0	89.4	31.6	88.4	31.2	31.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.0	5.1	4.4	20.1	35.4	2.3	1.8	18.5	36.7	17.1	37.8	15.9	38.8	14.2	40.2		
(5)			19.6	28.9	2.1	0.3	18.1	29.1	16.9	29.3	15.7	29.5	14.2	29.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	27.3	26.2	26.2	26.7	27.3	27.9	27.9	27.7	28.1	27.8	27.5	27.1	26.9
	DBStd	1.10	1.14	0.92	0.87	0.74	0.72	0.86	0.96	1.01	0.93	0.81	0.87	0.90
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6305	502	453	518	519	554	537	547	560	534	543	514	523
	CDD18.3	3263	243	220	259	269	296	287	289	302	284	284	264	265
	CDH23.3	33169	2059	1868	2448	2851	3336	3161	3057	3338	2987	2936	2629	2499
	CDH26.7	9662	351	329	558	859	1161	1076	998	1210	993	899	693	534
Wind	WSAvg	1.7	2.2	2.2	2.0	1.8	1.5	1.4	1.4	1.4	1.4	1.5	1.7	2.0
Precipitation	PrecAvg	4032	603	458	371	223	200	259	228	178	198	288	464	645
	PrecMax	6505	1171	1626	1264	531	416	450	409	391	338	490	860	1189
	PrecMin	2575	176	67	67	55	46	103	77	55	89	165	211	272
	PrecStd	1093	258	364	261	136	109	89	84	83	67	86	175	228
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	30.2	30.1	30.7	31.6	32.7	33.1	33.5	34.0	33.6	32.7	31.6	31.0
		MCWB	27.0	26.8	27.1	27.3	27.7	27.8	27.7	28.1	28.0	27.5	27.6	27.2
	2%	DB	29.6	29.5	30.1	30.9	31.7	32.0	32.2	32.9	32.2	31.6	30.8	30.2
		MCWB	26.7	26.6	26.8	27.3	27.7	27.8	27.5	27.7	27.5	27.3	27.3	27.0
	5%	DB	29.2	29.1	29.7	30.5	31.2	31.4	31.4	32.1	31.5	31.0	30.3	29.7
		MCWB	26.5	26.3	26.6	27.1	27.5	27.6	27.3	27.5	27.3	27.2	27.1	26.8
	10%	DB	28.6	28.6	29.2	30.0	30.7	30.8	30.7	31.2	30.7	30.3	29.8	29.1
		MCWB	26.2	26.0	26.4	26.9	27.3	27.4	27.1	27.2	27.1	27.0	26.9	26.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.6	27.5	27.7	28.0	28.2	28.5	28.2	28.6	28.3	28.2	28.0	27.7
		MCDB	29.7	29.4	29.9	30.6	31.5	31.5	31.6	32.7	32.2	31.4	30.7	30.1
	2%	WB	27.1	27.0	27.2	27.6	27.9	28.0	27.8	28.1	27.8	27.7	27.6	27.3
		MCDB	29.2	29.1	29.5	30.2	31.0	31.3	31.2	32.0	31.3	30.7	30.3	29.6
	5%	WB	26.7	26.6	26.9	27.4	27.7	27.7	27.5	27.7	27.6	27.4	27.3	27.1
		MCDB	28.7	28.6	29.3	30.0	30.6	30.8	30.8	31.3	30.9	30.3	29.8	29.3
	10%	WB	26.3	26.2	26.6	27.1	27.4	27.5	27.2	27.4	27.2	27.1	27.0	26.7
		MCDB	28.2	28.2	28.8	29.7	30.3	30.4	30.3	30.7	30.3	29.9	29.5	28.8
Mean Daily Temperature Range	5% DB	MDBR	4.2	4.6	5.1	5.6	5.9	6.0	6.0	6.6	6.3	5.9	5.2	4.4
		MCDBR	5.3	5.4	5.8	6.2	6.7	6.9	7.3	7.8	7.4	6.9	6.1	5.3
		MCWBR	3.3	3.4	3.5	3.5	3.5	3.5	3.6	3.7	3.6	3.5	3.4	3.0
	5% WB	MCDBR	4.9	5.1	5.5	5.9	6.4	6.5	6.7	7.2	6.9	6.5	5.8	5.1
		MCWBR	3.2	3.5	3.5	3.6	3.5	3.5	3.6	3.7	3.6	3.5	3.4	3.0
Clear-Sky Solar Irradiance	taub	0.477	0.451	0.443	0.431	0.427	0.422	0.424	0.424	0.446	0.437	0.449	0.450	0.472
	taud	2.242	2.333	2.378	2.418	2.455	2.480	2.471	2.391	2.414	2.382	2.368	2.269	
	Ebn at Noon	828	868	879	880	868	863	864	855	871	859	848	822	
	Edh at Noon	138	130	126	121	114	110	111	122	120	122	121	132	
All-Sky Solar Radiation	RadAvg	3.74	4.65	5.38	5.86	5.70	5.17	4.84	5.07	4.90	4.68	4.22	3.77	
	RadStd	0.54	0.59	0.58	0.42	0.42	0.32	0.38	0.31	0.45	0.33	0.40	0.32	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page